CASE STUDY

Automatic Parking System Using Counters and FSM

1. Introduction

AnAutomatic Parking System is a digital system designed to manage the entry, exit, and availability of vehicles in a parking area without requiring continuous manual supervision. The system uses counters to keep track of the number of vehicles inside the parking area and a Finite State Machine (FSM) to control the different operating states of the parking system.

In a conventional parking system, security personnel may have to manually count vehicles and check whether parking spaces are available. This can result in errors, delays, and inefficient use of parking space. A digital automatic parking system can solve these problems by continuously monitoring vehicle entry and exit.

The proposed system uses sensors to detect vehicles at the entrance and exit. The counter increases when a vehicle enters and decreases when a vehicle leaves. The FSM controls the sequence of operations, such as waiting for a vehicle, detecting entry, checking availability, opening the gate,and closing the gate.

2. Background

Adigital parking system can automatically determine whether parking spaces are available. For example, if a parking area has a capacity of 50 vehicles, a counter can maintain the current number of vehicles inside. When the count reaches 50, the system can indicate “Parking Full” and prevent additional vehicles from entering.

A counter is a sequential digital circuit that changes its output according to clock pulses or input events. In this application, the entry sensor can generate an event that increments the counter, while the exit sensor generates an event that decrements it.

An FSM is used because the parking system operates through a sequence of well-defined states. The FSM makes the control process organized and prevents incorrectoperations.

3. Problem Statement

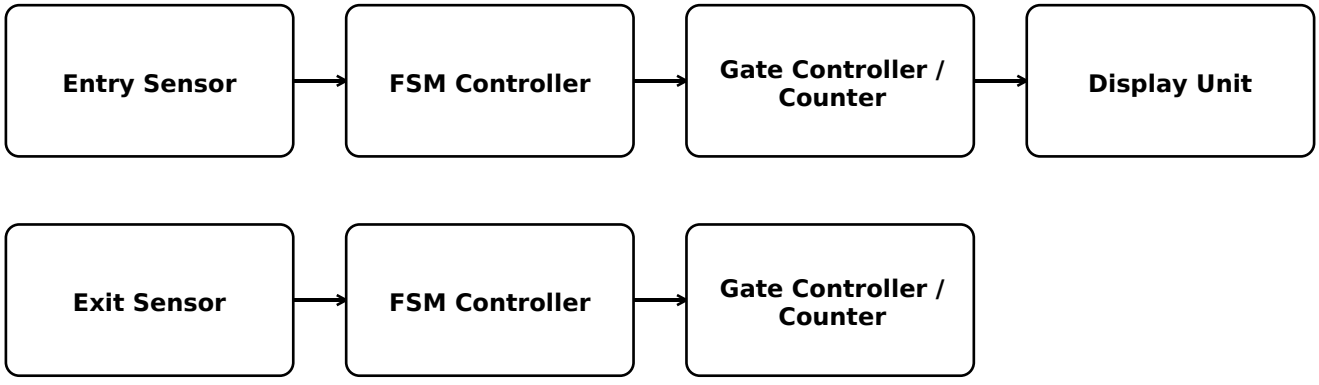
- Difficulty in knowing the exact numberof available spaces.
- Human errors while counting vehicles.
- Unnecessary waiting at parking entrances.
- Vehicles entering when the parking area is already full.
- Difficulty in automatically controlling entry and exit gates.
- Lack of real-time parking status information.

Therefore, there is a need for an automatic system that can detect vehicles, count them, control gates, and indicate parking availability. The proposed solution is an Automatic Parking System using Counters and Finite State Machines.

4. Objectives

- To automatically detect vehicle entry and exit.
- To maintain the number of vehicles currently inside the parking area.
- To use a counter for vehicle counting.
- To use an FSM to control the parking process.
- To automatically open and close the entrance and exit gates.
- To display the number of available parking spaces.
- To prevent vehicles from entering when the parking area is full.
- To reduce human involvement in parking management.
- To demonstrate the practical application of sequential digitalcircuits.

Basic Working Flow



5. Methodology

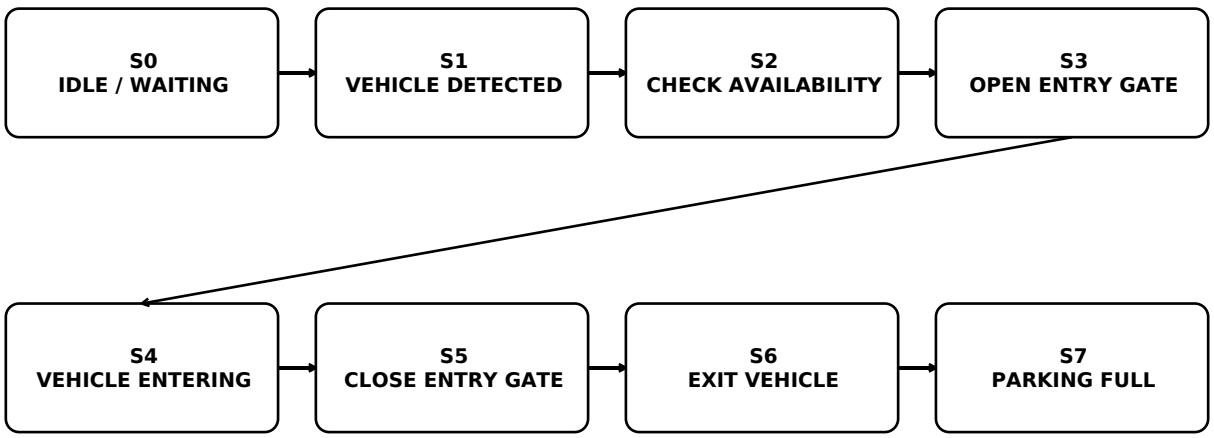
Entry Sensor → FSM Controller → Gate Controller / Counter → Display Unit

Exit Sensor → FSM Controller → Gate Controller / Counter When a vehicle approaches the entrance, the entry sensor detects it. The FSM checks the current parking status. lf a parking space is available, the FSM changes to the Entry state and sends a signal to open the entrance gate. After the vehicle enters, the counter is incremented by one.

When a vehicle approaches the exit, the exit sensor detects it. The FSM changes to the Exit state, opens the exit gate, and the counter is decremented after the vehicle leaves. The display continuously shows the current number of vehicles or available spaces.

The FSMcan be designed using different states according to the requirements of the parking system. Example states: S0 – Idle/Waiting; S1 – Vehicle Detected; S2 – Check Availability; S3 – Open Entry Gate; S4 – Vehicle Entering; S5 – Close Entry Gate; S6 – Exit Vehicle; S7 – Parking Full.

FSM State Flow



FSM controls the sequence and prevents entry when capacity is reached.

6. Analysis and Interpretation

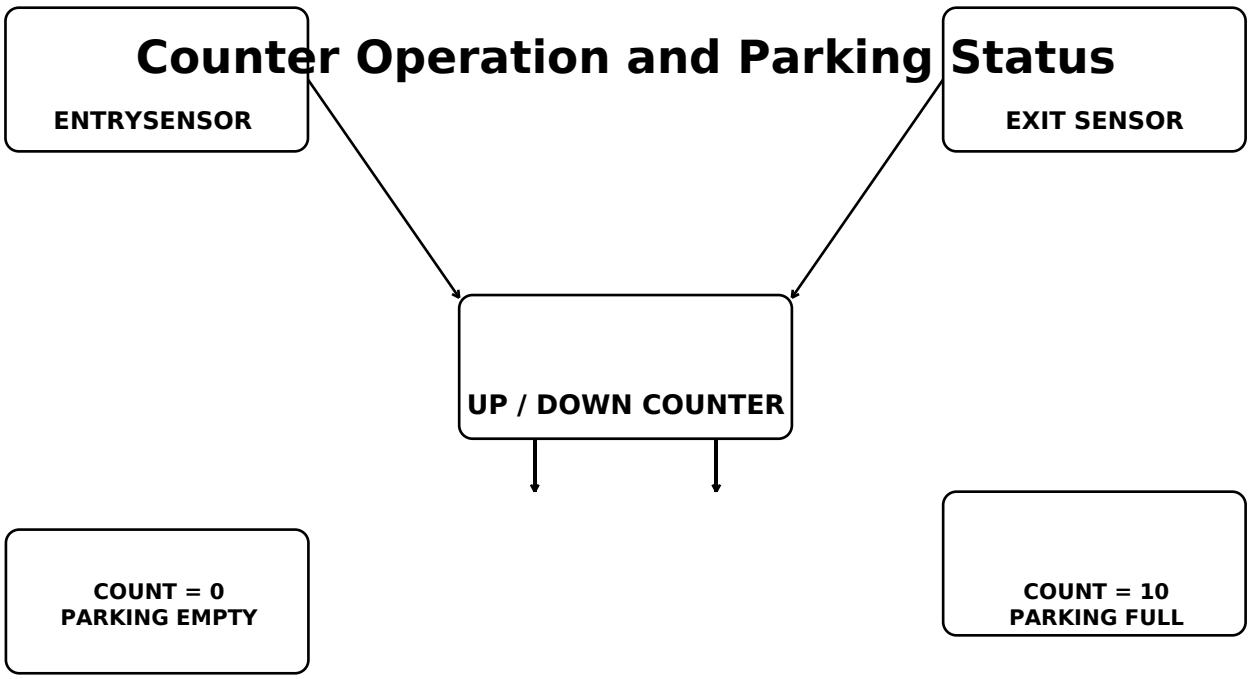
Example FSM flow: IDLE → VEHICLEDETECTED → CHECK

AVAILABILITY → OPEN GATE → VEHICLE ENTERS → INCREMENT COUNTER → CLOSE GATE → IDLE

The counter maintains the number of vehicles inside the parking area. For a parking capacity of 10 vehicles: Vehicle enters → Counter = Counter + 1; Vehicle leaves → Counter = Counter − 1; Counter = 0 → Parking is empty; Counter = 10 → Parking is full.

Available Spaces = Maximum Capacity − Current Vehicles. For example, if maximum capacity is 10 and current vehicles are 7, available spaces are 3.

The FSM ensures that the entrance gate does not open when the counter reaches the maximum capacity.



Available Spaces = Maximum Capacity − Current Vehicles

For capacity 10: count 0 = empty; count 10 = full.

7. Findings

The case studyshows that counters and FSMs can effectively be used to develop an automatic parking management system.

- Vehicle counting can be automated.
- Entry and exit can be monitored continuously.
- The system can identify whether parking is full.
- FSM provides systematic control of the parking sequence.
- Counters provide accurate vehicle-count information.
- Human intervention can be reduced.
- The system can provide real-time parking information.
- The same concept can be expanded for larger parking facilities.

8. Discussion

Thecombinationof a counter and FSM is useful because both components perform different functions. The counter is mainly responsible for maintaining numerical information about the vehicles inside the parking area. The FSM is responsible for controlling the sequence of operations.

Simply using a counter cannot decide when the gate should open or close. Similarly, an FSM alone is not suitable for maintaining the total number of vehicles. Combining both creates a more complete control system.

The system can also be connected to LEDs or a seven-segment display to show parking availability. In a more advanced implementation, LCD displays or digital signboards can be used.

9. Recommendations

- Use reliableIR,ultrasonic, or othersuitable vehicle-detection sensors.
- Add a seven-segment or LCD display for available-space information.
- Add separate indicators for Parking Available and Parking Full.
- Use a microcontroller or FPGA for a larger implementation.
- Add RFID or QR-based vehicle identification.
- Add multiple entry and exit gates for large parking areas.
- Include protection against incorrect counter operation.
- Add a mobile or web interface for real-time parking information.
- Include emergency gate-opening functionality.
- Add individual parking-space sensors for detailed space-level monitoring.

10. Conclusion

TheAutomatic Parking System Using Counters and FSM demonstrates a practical application of digital electronics and sequential logic. The counter maintains the number of vehicles in the parking area, while the FSM controls the sequence of entry and exit operations.

The system can automatically detect vehicles, control gates, update the vehicle count, and indicate whether parking spaces are available. The concept can be implemented using basic digital logic circuits and can later be expanded using microcontrollers, FPGAs, sensors, RFID systems, and IoT technologies.

Thus, the proposed system provides a useful example of how counters and finite state machines can be combined to solve a real-world automation problem.